INFOSYS-STYLE PSEUDOCODE PRACTICE

100 MEDIUM-LEVEL MCQs • TOPIC-WISE • ANSWERS & SOLUTIONS

TOPICS COVERED

No.	Topic	Coverage
1	**Operators & Precedence**	Arithmetic, relational, logical, pre/post increment and decrement
2	**Bitwise Operators**	AND (&), OR (|), XOR (^) and mixed bitwise expressions
3	**Left & Right Shift**	Left shift (<<), right shift (>>) and shift-based expressions
4	**Loops**	For, while, reverse loops, changing state and loop conditions
5	**Nested Loops**	Nested iteration, counting, sums, conditions and break
6	**Break & Continue**	Control flow inside loops
7	**Arrays**	Indexing, updates, running sums, comparisons, nested indexing
8	**Strings**	Length, indexing, character checks and loops
9	**Functions**	Parameters, return values, loops, bitwise and nested functions
10	**Mixed Logic**	Placement-style combinations of arrays, loops, functions, shifts and conditions

Difficulty: Medium. Questions are designed to require 2–5 reasoning steps rather than direct one-line outputs.
Notation: ^ = XOR, & = AND, | = OR, << = left shift, >> = right shift, % = remainder. Arrays and strings use 0-based indexing. Integer division discards decimals.

Topic: Operator Precedence

Q1. What will be the output of the following pseudocode?

```
Set Integer a = 18
Set Integer b = 5
Set Integer c = 2
display a - b * c + 6
```

A. 10

B. 14

C. 16

D. 19

Topic: Mixed Arithmetic

Q2. What will be the output of the following pseudocode?

```
Set Integer x = 24
Set Integer y = 7
display (x / y) * y + (x % y)
```

A. 21

B. 24

C. 25

D. 28

Topic: Logical Operators

Q3. What will be the output of the following pseudocode?

```
Set Integer a = 12
Set Integer b = 7
Set Integer c = 5
display (a > b) and ((b - c) == 2)
```

A. true

B. false

C. 2

D. 12

Topic: Logical Operators

Q4. What will be the output of the following pseudocode?

```
Set Integer x = 9
display (x % 2 == 0) or ((x > 5) and (x < 10))
```

A. true

B. false

C. 9

D. 0

Topic: Bitwise XOR

Q5. What will be the output of the following pseudocode?

```
Set Integer a = 15
Set Integer b = 9
Set Integer c = 6
display (a ^ b) + c
```

A. 8

B. 10

C. 12

D. 14

Topic: Bitwise AND

Q6. What will be the output of the following pseudocode?

```
Set Integer a = 27
Set Integer b = 14
display (a & b) + (a % b)
```

A. 19

B. 21

C. 23

D. 25

Topic: Bitwise OR

Q7. What will be the output of the following pseudocode?

```
Set Integer a = 18
Set Integer b = 11
display (a | b) - (a & b)
```

A. 21

B. 23

C. 25

D. 27

Topic: XOR + AND

Q8. What will be the output of the following pseudocode?

```
Set Integer a = 13
Set Integer b = 7
display (a ^ b) + (a & b)
```

A. 13

B. 15

C. 17

D. 19

Topic: Pre/Post Increment

Q9. What will be the output of the following pseudocode?

```
Set Integer x = 4
Set Integer y = x++ + ++x
Set Integer z = y + x
display z
```

A. 14

B. 15

C. 16

D. 17

Topic: Post/Pre Decrement

Q10. What will be the output of the following pseudocode?

```
Set Integer x = 10
Set Integer y = --x
Set Integer z = y + x--
display z
```

A. 16

B. 17

C. 18

D. 19

Topic: Relational Logic

Q11. What will be the output of the following pseudocode?

```
Set Integer p = 8
Set Integer q = 12
display (p + q > 20) and (q - p == 4)
```

A. true

B. false

C. 4

D. 20

Topic: Ternary Logic

Q12. What will be the output of the following pseudocode?

```
Set Integer a = 14
Set Integer b = 9
Set Integer m = (a % 2 == 0) ? a/2 : b*2
display m + b
```

A. 14

B. 16

C. 18

D. 21

Topic: Compound Assignment

Q13. What will be the output of the following pseudocode?

```
Set Integer x = 7
x = x * 3 - 4
x = x + (x % 5)
display x
```

A. 18

B. 19

C. 20

D. 21

Topic: Mixed Operators

Q14. What will be the output of the following pseudocode?

```
Set Integer a = 20
Set Integer b = 6
display (a / b) + (a % b) * 2
```

A. 5

B. 7

C. 9

D. 11

Topic: Left Shift

Q15. What will be the output of the following pseudocode?

```
Set Integer x = 11
Set Integer y = 2
display (x << y) - x
```

A. 22

B. 33

C. 44

D. 55

Topic: Right Shift

Q16. What will be the output of the following pseudocode?

```
Set Integer x = 45
Set Integer y = 2
display (x >> y) + x % 4
```

A. 10

B. 11

C. 12

D. 13

Topic: Left Shift in Expression

Q17. What will be the output of the following pseudocode?

```
Set Integer a = 6
Set Integer b = 3
display (a << b) / 4
```

A. 8

B. 10

C. 12

D. 16

Topic: Right Shift in Expression

Q18. What will be the output of the following pseudocode?

```
Set Integer a = 50
display (a >> 1) - (a >> 3)
```

A. 17

B. 19

C. 21

D. 23

Topic: Shift + XOR

Q19. What will be the output of the following pseudocode?

```
Set Integer x = 9
display (x << 1) ^ x
```

A. 19

B. 23

C. 27

D. 31

Topic: Shift + AND

Q20. What will be the output of the following pseudocode?

```
Set Integer x = 29
display (x >> 2) & 7
```

A. 5

B. 6

C. 7

D. 8

Topic: Double Shift

Q21. What will be the output of the following pseudocode?

```
Set Integer x = 7
Set Integer y = (x << 2)
display y >> 1
```

A. 7

B. 12

C. 14

D. 28

Topic: Shift and Modulo

Q22. What will be the output of the following pseudocode?

```
Set Integer x = 37
display (x >> 1) % 6
```

A. 0

B. 1

C. 2

D. 3

Topic: Shift Condition

Q23. What will be the output of the following pseudocode?

```
Set Integer x = 23
if ((x >> 2) >= 5)
display x - 3
else
display x + 3
end-if
```

A. 20

B. 23

C. 26

D. 28

Topic: Shift + Arithmetic

Q24. What will be the output of the following pseudocode?

```
Set Integer a = 14
Set Integer b = 2
display (a << b) - (a >> b)
```

A. 51

B. 53

C. 55

D. 57

Topic: Shift Chain

Q25. What will be the output of the following pseudocode?

```
Set Integer x = 3
x = x << 2
x = x >> 1
x = x << 1
display x
```

A. 6

B. 10

C. 12

D. 24

Topic: Shift + OR

Q26. What will be the output of the following pseudocode?

```
Set Integer x = 5
display (x << 2) | 3
```

A. 19
B. 20
C. 23
D. 24

Topic: Shift + XOR

Q27. What will be the output of the following pseudocode?

```
Set Integer x = 12
display (x >> 1) ^ 5
```

A. 1
B. 3
C. 5
D. 7

Topic: Bit Test with Shift

Q28. What will be the output of the following pseudocode?

```
Set Integer x = 40
display (x >> 3) & 1
```

A. 0
B. 1
C. 2
D. 5

Topic: Shifted Difference

Q29. What will be the output of the following pseudocode?

```
Set Integer a = 31
display (a << 1) - (a >> 2)
```

A. 54
B. 55
C. 56
D. 57

Topic: For Loop with Condition

Q30. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 10 step 1 do
if (i % 3 != 0)
s = s + i
end-if
end-for
display s
```

A. 30
B. 37
C. 42
D. 45

Topic: Nested Loop

Q31. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 5 step 1 do
if (i + j == 6)
c = c + 1
end-if
end-for
end-for
display c
```

A. 2

B. 3

C. 4

D. 5

Topic: Nested Loop Sum

Q32. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to i + 1 step 1 do
s = s + j
end-for
end-for
display s
```

A. 17

B. 18

C. 19

D. 20

Topic: While Loop

Q33. What will be the output of the following pseudocode?

```
Set Integer x = 5
Set Integer s = 0
while (x <= 25)
s = s + x
x = x + 5
end-while
display s
```

A. 50

B. 60

C. 75

D. 80

Topic: Break in Nested Loop

Q34. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if (j == 3)
break
end-if
c = c + 1
end-for
end-for
display c
```

A. 4

B. 6

C. 8

D. 12

Topic: Continue

Q35. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 8 step 1 do
if (i % 3 == 0)
continue
end-if
s = s + i
end-for
display s
```

A. 24

B. 27

C. 30

D. 33

Topic: Reverse Loop

Q36. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 12 to 2 step -2 do
s = s + i / 2
end-for
display s
```

A. 18

B. 21

C. 24

D. 27

Topic: Loop with Changing Step

Q37. What will be the output of the following pseudocode?

```
Set Integer i = 1
Set Integer s = 0
while (i < 20)
s = s + i
i = i * 2 + 1
end-while
display s
```

A. 24

B. 26

C. 28

D. 30

Topic: Nested Loop Count

Q38. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 5 step 1 do
for j = i to 5 step 1 do
c = c + 1
end-for
end-for
display c
```

A. 10

B. 12

C. 15

D. 20

Topic: Loop with Modulo

Q39. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 14 step 2 do
if (i % 4 == 0)
s = s + i
end-if
end-for
display s
```

A. 12

B. 16

C. 20

D. 24

Topic: Nested Conditional Loop

Q40. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 5 step 1 do
if (i % 2 == 1)
for j = 1 to i step 2 do
s = s + j
end-for
end-if
end-for
display s
```

A. 12

B. 14

C. 16

D. 18

Topic: Loop State

Q41. What will be the output of the following pseudocode?

```
Set Integer x = 3
for i = 1 to 4 step 1 do
x = x + i
x = x - 1
end-for
display x
```

A. 8

B. 9

C. 10

D. 11

Topic: Loop Product

Q42. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 2 to 8 step 2 do
p = p * (i / 2)
end-for
display p
```

A. 12

B. 24

C. 36

D. 48

Topic: While + Modulo

Q43. What will be the output of the following pseudocode?

```
Set Integer n = 29
Set Integer c = 0
while (n > 0)
if (n % 2 == 1)
c = c + 1
end-if
n = n / 2
end-while
display c
```

A. 2

B. 3

C. 4

D. 5

Topic: Nested Loop Arithmetic

Q44. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to 3 step 1 do
s = s + i * j
end-for
end-for
display s
```

A. 12

B. 18

C. 24

D. 36

Topic: Break with Sum

Q45. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 12 step 2 do
s = s + i
if (s > 15)
break
end-if
end-for
display s
```

A. 12

B. 18

C. 20

D. 24

Topic: Continue with Product

Q46. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 1 to 6 step 1 do
if (i % 2 == 0)
continue
end-if
p = p * i
end-for
display p
```

A. 9

B. 12

C. 15

D. 20

Topic: Loop with Shift

Q47. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 4 step 1 do
s = s + (i << 1)
end-for
display s
```

A. 16

B. 18

C. 20

D. 24

Topic: Loop with Right Shift

Q48. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 8 to 32 step 8 do
s = s + (i >> 2)
end-for
display s
```

A. 15

B. 18

C. 20

D. 24

Topic: Nested Loop with Break

Q49. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
c = c + 1
if (c == 7)
break
end-if
end-for
end-for
display c
```

A. 7

B. 8

C. 12

D. 13

Topic: Loop and Comparison

Q50. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 9 step 1 do
if (i * i < 50)
c = c + 1
end-if
end-for
display c
```

A. 6

B. 7

C. 8

D. 9

Topic: Array Index Calculation

Q51. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 2, 9, 5]
display a[a[2]] + a[1]
```

A. 7

B. 9

C. 11

D. 13

Topic: Array Transformation

Q52. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
for i = 0 to 3 step 1 do
a[i] = a[i] + i
end-for
display a[3] - a[1]
```

A. 3

B. 4

C. 5

D. 6

Topic: Array Maximum

Q53. What will be the output of the following pseudocode?

```
Set Integer a = [8, 3, 15, 6, 11]
Set Integer m = a[0]
for i = 1 to 4 step 1 do
if (a[i] > m)
m = a[i]
end-if
end-for
display m % 4
```

A. 1

B. 2

C. 3

D. 4

Topic: Array Pair Sum

Q54. What will be the output of the following pseudocode?

```
Set Integer a = [2, 5, 8, 11]
Set Integer s = 0
for i = 0 to 2 step 1 do
s = s + a[i] + a[i+1]
end-for
display s
```

A. 36

B. 39

C. 42

D. 45

Topic: Array Conditional Update

Q55. What will be the output of the following pseudocode?

```
Set Integer a = [4, 7, 10, 13]
for i = 0 to 3 step 1 do
if (a[i] % 2 == 0)
a[i] = a[i] / 2
else
a[i] = a[i] + 1
end-if
end-for
display a[0] + a[1] + a[2] + a[3]
```

A. 25

B. 27

C. 29

D. 31

Topic: Array Reverse

Q56. What will be the output of the following pseudocode?

```
Set Integer a = [3, 6, 9, 12, 15]
display a[4] - a[0] + a[3] - a[1]
```

A. 12

B. 15

C. 18

D. 21

Topic: Array Count

Q57. What will be the output of the following pseudocode?

```
Set Integer a = [5, 10, 15, 20, 25]
Set Integer c = 0
for i = 0 to 4 step 1 do
if (a[i] % 10 == 5)
c = c + 1
end-if
end-for
display c
```

A. 2

B. 3

C. 4

D. 5

Topic: Array Running Sum

Q58. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8]
Set Integer s = a[0]
for i = 1 to 3 step 1 do
a[i] = a[i] + s
s = a[i]
end-for
display a[3]
```

A. 16

B. 20

C. 24

D. 30

Topic: Array Nested Index

Q59. What will be the output of the following pseudocode?

```
Set Integer a = [1, 3, 0, 2]
display a[a[3]] + a[a[1]]
```

A. 1

B. 2

C. 3

D. 4

Topic: Array Difference

Q60. What will be the output of the following pseudocode?

```
Set Integer a = [12, 5, 9, 4]
Set Integer m = a[0]
for i = 1 to 3 step 1 do
if (a[i] < m)
m = a[i]
end-if
end-for
display a[0] - m
```

A. 6

B. 7

C. 8

D. 9

Topic: Array Shift

Q61. What will be the output of the following pseudocode?

```
Set Integer a = [3, 5, 7, 9]
display (a[3] >> 1) + (a[0] << 1)
```

A. 8

B. 10

C. 12

D. 14

Topic: Array XOR

Q62. What will be the output of the following pseudocode?

```
Set Integer a = [6, 10, 12, 3]
display (a[0] ^ a[1]) + a[3]
```

A. 13

B. 15

C. 17

D. 19

Topic: Array Pair Comparison

Q63. What will be the output of the following pseudocode?

```
Set Integer a = [4, 9, 6, 11]
Set Integer c = 0
for i = 0 to 2 step 1 do
if (a[i] < a[i+1])
c = c + 1
end-if
end-for
display c
```

A. 1

B. 2

C. 3

D. 4

Topic: Array Sum with Step

Q64. What will be the output of the following pseudocode?

```
Set Integer a = [2, 4, 6, 8, 10, 12]
Set Integer s = 0
for i = 0 to 5 step 2 do
s = s + a[i]
end-for
display s
```

A. 12

B. 16

C. 18

D. 20

Topic: Array Rotation Logic

Q65. What will be the output of the following pseudocode?

```
Set Integer a = [10, 20, 30, 40]
Set Integer t = a[3]
for i = 3 to 1 step -1 do
a[i] = a[i-1]
end-for
a[0] = t
display a[1] + a[3]
```

A. 30

B. 40

C. 50

D. 60

Topic: String Length

Q66. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer n = stringLength(s)
display (n << 1) - 3
```

A. 13

B. 15

C. 17

D. 19

Topic: String Indexing

Q67. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[1] + s[5]
```

A. NS

B. NY

C. FY

D. SY

Topic: String Condition

Q68. What will be the output of the following pseudocode?

```
Set String s = "COMPUTER"
if (stringLength(s) % 2 == 0)
display stringLength(s) / 2
else
display stringLength(s) + 2
end-if
```

A. 4

B. 5

C. 6

D. 10

Topic: String + Loop

Q69. What will be the output of the following pseudocode?

```
Set String s = "CODE"
Set Integer total = 0
for i = 0 to stringLength(s)-1 step 1 do
total = total + i
end-for
display total
```

A. 4

B. 5

C. 6

D. 7

Topic: String Index + Shift

Q70. What will be the output of the following pseudocode?

```
Set String s = "TESTING"
Set Integer n = s[1] == 'E'
display stringLength(s) << n
```

A. 7

B. 14

C. 21

D. 28

Topic: String Length + Modulo

Q71. What will be the output of the following pseudocode?

```
Set String s = "ALGORITHM"
display stringLength(s) % 4
```

A. 0

B. 1

C. 2

D. 3

Topic: String Reverse Access

Q72. What will be the output of the following pseudocode?

```
Set String s = "INFOSYS"
display s[stringLength(s)-1]
```

A. I

B. S

C. Y

D. Error

Topic: String Character Count

Q73. What will be the output of the following pseudocode?

```
Set String s = "PLACEMENT"
Set Integer c = 0
for i = 0 to stringLength(s)-1 step 1 do
if (s[i] == 'E')
c = c + 1
end-if
end-for
display c
```

A. 1

B. 2

C. 3

D. 4

Topic: String Length + Right Shift

Q74. What will be the output of the following pseudocode?

```
Set String s = "INTERVIEW"
display stringLength(s) >> 1
```

A. 3

B. 4

C. 5

D. 6

Topic: String Conditional

Q75. What will be the output of the following pseudocode?

```
Set String s = "NQT2026"
if (stringLength(s) > 6)
display s[0]
else
display s[1]
end-if
```

A. N

B. Q

C. T

D. 2

Topic: Function Arithmetic

Q76. What will be the output of the following pseudocode?

```
Integer calc(Integer a, Integer b)
a = a + b
b = a - b
return a * b
End function
display calc(7, 3)
```

A. 60

B. 70

C. 80

D. 90

Topic: Function with Loop

Q77. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 1
for i = 1 to n step 1 do
s = s + i
end-for
return s
End function
display f(6)
```

A. 20

B. 21

C. 22

D. 23

Topic: Function with Shift

Q78. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return (n << 2) + (n >> 1)
End function
display f(10)
```

A. 40

B. 45

C. 50

D. 55

Topic: Function with XOR

Q79. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
return (a ^ b) * 2
End function
display f(12, 5)
```

A. 18

B. 26

C. 30

D. 34

Topic: Function Condition

Q80. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n % 4 == 0)
return n / 2
else
return n + 4
end-if
End function
display f(14)
```

A. 9

B. 14

C. 18

D. 28

Topic: Function Digit Logic

Q81. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
while (n > 0)
s = s + n % 10
n = n / 10
end-while
return s
End function
display f(386)
```

A. 14

B. 15

C. 16

D. 17

Topic: Function Nested

Q82. What will be the output of the following pseudocode?

```
Integer g(Integer x)
return x * 2
End function

Integer f(Integer x)
return g(x) + g(x+1)
End function

display f(4)
```

A. 16

B. 18

C. 20

D. 22

Topic: Function Modulo

Q83. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return (n * n + 3) % 10
End function
display f(7)
```

A. 2

B. 5

C. 8

D. 9

Topic: Function Loop Product

Q84. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer p = 1
for i = 1 to n step 1 do
if (i % 2 == 0)
p = p * i
end-if
end-for
return p
End function
display f(6)
```

A. 24

B. 36

C. 48

D. 60

Topic: Function Boolean

Q85. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
if ((a & 1) == 1 and (b & 1) == 0)
return a + b
else
return a - b
end-if
End function
display f(9, 6)
```

A. 3

B. 9

C. 15

D. 54

Topic: Function Array Formula

Q86. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b, Integer c)
return (a + b) ^ c
End function
display f(5, 3, 6)
```

A. 4

B. 8

C. 12

D. 14

Topic: Function Recursive-style

Q87. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n <= 1)
return 1
end-if
return n + f(n-2)
End function
display f(7)
```

A. 10

B. 12

C. 16

D. 18

Topic: Function Shifted Return

Q88. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer a = n << 1
Set Integer b = n >> 1
return a - b
End function
display f(13)
```

A. 13

B. 19

C. 20

D. 21

Topic: Function with Multiple Updates

Q89. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
a = a * 2
b = b + a
a = a - b
return a + b
End function
display f(4, 3)
```

A. 7

B. 8

C. 9

D. 10

Topic: Function with Loop and Shift

Q90. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
for i = 1 to n step 1 do
s = s + (i << 1)
end-for
return s
End function
display f(5)
```

A. 20

B. 25

C. 30

D. 35

Topic: Mixed Array + Function

Q91. What will be the output of the following pseudocode?

```
Integer f(Integer x)
return (x >> 1) + 3
End function

Set Integer a = [8, 12, 16]
display f(a[2])
```

A. 9

B. 10

C. 11

D. 12

Topic: Mixed Loop + XOR

Q92. What will be the output of the following pseudocode?

```
Set Integer x = 0
for i = 1 to 5 step 1 do
x = x ^ i
end-for
display x
```

A. 0

B. 1

C. 2

D. 3

Topic: Mixed Loop + AND

Q93. What will be the output of the following pseudocode?

```
Set Integer x = 15
for i = 1 to 3 step 1 do
x = x & (14 - i)
end-for
display x
```

A. 8

B. 10

C. 12

D. 14

Topic: Mixed Condition + Shift

Q94. What will be the output of the following pseudocode?

```
Set Integer x = 19
if ((x & 1) == 1)
x = x << 1
x = x + 3
else
x = x >> 1
end-if
display x
```

A. 19

B. 38

C. 41

D. 44

Topic: Mixed Nested Loop + Modulo

Q95. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if ((i + j) % 3 == 0)
c = c + 1
end-if
end-for
end-for
display c
```

A. 4

B. 5

C. 6

D. 7

Topic: Mixed Array + Right Shift

Q96. What will be the output of the following pseudocode?

```
Set Integer a = [18, 25, 32, 41]
display (a[3] >> 2) + (a[0] % 5)
```

A. 11

B. 12

C. 13

D. 14

Topic: Mixed Function + Logical

Q97. What will be the output of the following pseudocode?

```
Integer f(Integer a, Integer b)
if (a > 10 and b < 10)
return (a << 1) + b
else
return a + (b << 1)
end-if
End function
display f(12, 7)
```

A. 24

B. 26

C. 31

D. 38

Topic: Mixed Loop + Array Update

Q98. What will be the output of the following pseudocode?

```
Set Integer a = [1, 2, 3, 4, 5]
for i = 1 to 4 step 1 do
a[i] = a[i] + a[i-1]
end-for
display a[4]
```

A. 10

B. 12

C. 15

D. 16

Topic: Mixed Function + Modulo + Shift

Q99. What will be the output of the following pseudocode?

```
Integer f(Integer n)
return ((n % 5) << 2) + (n >> 3)
End function
display f(26)
```

A. 5

B. 7

C. 9

D. 11

Topic: Mixed Nested Conditions

Q100. What will be the output of the following pseudocode?

```
Set Integer a = 9
Set Integer b = 6
Set Integer c = 3
if (a > b)
if ((a - b) == c)
display a + b + c
else
display a - b
end-if
else
display b + c
end-if
```

A. 12

B. 15

C. 18

D. 21

ANSWER KEY & SOLUTIONS

Each solution shows the key calculation or logic needed to reach the answer.

Q1. Answer: B — 14

Topic: Operator Precedence
Multiplication first: 18 – (5×2) + 6 = 14.

Q2. Answer: B — 24

Topic: Mixed Arithmetic
24/7=3 and 24%7=3; 3×7+3 = 24.

Q3. Answer: A — true

Topic: Logical Operators
12>7 is true and 7–5=2 is true, so the result is true.

Q4. Answer: A — true

Topic: Logical Operators
First condition is false, but 9>5 and 9<10 are both true.

Q5. Answer: C — 12

Topic: Bitwise XOR
15^9 = 6; 6+6 = 12.

Q6. Answer: C — 23

Topic: Bitwise AND
27&14 = 10 and 27%14 = 13; total = 23.

Q7. Answer: C — 25

Topic: Bitwise OR
18|11=27 and 18&11=2; 27–2=25.

Q8. Answer: B — 15

Topic: XOR + AND
13^7=10 and 13&7=5; total = 15.

Q9. Answer: C — 16

Topic: Pre/Post Increment
x++ contributes 4 and makes x=5; ++x makes x=6 and contributes 6. y=10; z=10+6=16.

Q10. Answer: C — 18

Topic: Post/Pre Decrement
--x makes x=9, y=9; x-- contributes 9, then x becomes 8. z=18.

Q11. Answer: B — false

Topic: Relational Logic
p+q=20, so 20>20 is false. The second condition is true; false AND true = false.

Q12. Answer: B — 16

Topic: Ternary Logic
a is even, so m=7; 7+9=16.

Q13. Answer: B — 19

Topic: Compound Assignment
7×3–4=17; 17%5=2; final x=19.

Q14. Answer: B — 7

Topic: Mixed Operators
20/6=3; 20%6=2; 3+2×2=7.

Q15. Answer: B — 33

Topic: Left Shift
11<<2 = 44; 44–11=33.

Q16. Answer: C — 12

Topic: Right Shift
45>>2 = 11; 45%4=1; total=12.

Q17. Answer: C — 12

Topic: Left Shift in Expression
6<<3=48; 48/4=12.

Q18. Answer: B — 19

Topic: Right Shift in Expression
50>>1=25 and 50>>3=6; 25–6=19.

Q19. Answer: C — 27

Topic: Shift + XOR
9<<1=18; 18 XOR 9 = 27.

Q20. Answer: C — 7

Topic: Shift + AND
29>>2=7; 7 AND 7 = 7.

Q21. Answer: C — 14

Topic: Double Shift
7<<2=28; 28>>1=14.

Q22. Answer: A — 0

Topic: Shift and Modulo
37>>1=18; 18%6=0.

Q23. Answer: A — 20

Topic: Shift Condition
23>>2=5, so the condition is true and 23–3=20.

Q24. Answer: B — 53

Topic: Shift + Arithmetic
14<<2=56; 14>>2=3; 56–3=53.

Q25. Answer: C — 12

Topic: Shift Chain
3→12→6→12.

Q26. Answer: C — 23

Topic: Shift + OR
5<<2=20; 20|3=23.

Q27. Answer: B — 3

Topic: Shift + XOR
12>>1=6; 6 XOR 5 = 3.

Q28. Answer: B — 1

Topic: Bit Test with Shift
40>>3=5; 5 AND 1 = 1.

Q29. Answer: B — 55

Topic: Shifted Difference
31<<1=62; 31>>2=7; 62−7=55.

Q30. Answer: B — 37

Topic: For Loop with Condition
Total 1..10=55; multiples of 3 are 3+6+9=18; 55−18=37.

Q31. Answer: C — 4

Topic: Nested Loop
Pairs are (1,5),(2,4),(3,3),(4,2): 4.

Q32. Answer: C — 19

Topic: Nested Loop Sum
For i=1,2,3 the inner sums are 3,6,10; total=19.

Q33. Answer: C — 75

Topic: While Loop
5+10+15+20+25=75.

Q34. Answer: C — 8

Topic: Break in Nested Loop
The inner loop adds 1 for j=1 and j=2, then breaks at j=3. 4×2=8.

Q35. Answer: B — 27

Topic: Continue
1..8 sum=36; skip 3 and 6, so 36−9=27.

Q36. Answer: B — 21

Topic: Reverse Loop
Values of i/2: 6,5,4,3,2,1; sum=21.

Q37. Answer: B — 26

Topic: Loop with Changing Step
i values are 1,3,7,15; sum=26.

Q38. Answer: C — 15

Topic: Nested Loop Count
Counts are 5+4+3+2+1=15.

Q39. Answer: D — 24

Topic: Loop with Modulo
Multiples of 4 are 4,8,12; sum=24.

Q40. Answer: B — 14

Topic: Nested Conditional Loop
For i=1,3,5 the inner sums are 1,4,9; total=14.

Q41. Answer: B — 9

Topic: Loop State
Each iteration changes x by i−1. Total change=0+1+2+3=6; final x=9.

Q42. Answer: B — 24

Topic: Loop Product
Factors are 1,2,3,4; product=24.

Q43. Answer: B — 3

Topic: While + Modulo
Binary 29 is 11101, containing four 1s.

Q44. Answer: D — 36

Topic: Nested Loop Arithmetic
The double sum factorizes: (1+2+3)×(1+2+3)=6×6=36.

Q45. Answer: B — 18

Topic: Break with Sum
Running sums: 2,6,12,20; break after 20.

Q46. Answer: C — 15

Topic: Continue with Product
Odd values 1,3,5 are multiplied: 1×3×5=15.

Q47. Answer: C — 20

Topic: Loop with Shift
2+4+6+8=20.

Q48. Answer: C — 20

Topic: Loop with Right Shift
The shifted values are 2,4,6,8; sum=20.

Q49. Answer: D — 13

Topic: Nested Loop with Break
The counter continues across outer iterations. It reaches 7, breaks the current inner loop, then continues; 4 more inner iterations occur, so final c=13.

Q50. Answer: B — 7

Topic: Loop and Comparison
i=1..7 satisfy $i^2<50$; count=7.

Q51. Answer: B — 9

Topic: Array Index Calculation
a[2]=2, so a[a[2]]=a[2]=2; 2+7=9.

Q52. Answer: D — 6

Topic: Array Transformation
After updates, a=[3,6,9,12]; 12–6=6.

Q53. Answer: C — 3

Topic: Array Maximum
Maximum is 15; 15%4=3.

Q54. Answer: B — 39

Topic: Array Pair Sum
Pairs: 7+13+19=39.

Q55. Answer: C — 29

Topic: Array Conditional Update
Array becomes [2,8,5,14]; sum=29.

Q56. Answer: C — 18

Topic: Array Reverse
15–3+12–6=18.

Q57. Answer: B — 3

Topic: Array Count
5,15,25 satisfy the condition: 3.

Q58. Answer: B — 20

Topic: Array Running Sum
Running values become 2,6,12,20; a[3]=20.

Q59. Answer: B — 2

Topic: Array Nested Index
a[3]=2, so a[2]=0; a[1]=3, so a[3]=2; total=2.

Q60. Answer: C — 8

Topic: Array Difference
Minimum is 4; 12–4=8.

Q61. Answer: B — 10

Topic: Array Shift
9>>1=4; 3<<1=6; total=10.

Q62. Answer: B — 15

Topic: Array XOR
6^10=12; 12+3=15.

Q63. Answer: B — 2

Topic: Array Pair Comparison
The true comparisons are 4<9 and 6<11, so count=2.

Q64. Answer: C — 18

Topic: Array Sum with Step
Indices 0,2,4 give 2+6+10=18.

Q65. Answer: B — 40

Topic: Array Rotation Logic
After right rotation: [40,10,20,30]; a[1]+a[3]=40.

Q66. Answer: B — 15

Topic: String Length
Length=9; 9<<1=18; 18–3=15.

Q67. Answer: B — NY

Topic: String Indexing
s[1]=N and s[5]=Y, so the result is NY.

Q68. Answer: A — 4

Topic: String Condition
COMPUTER has 8 characters; 8/2=4.

Q69. Answer: C — 6

Topic: String + Loop
Indices 0+1+2+3=6.

Q70. Answer: B — 14

Topic: String Index + Shift
The comparison is true, represented as 1; length 7<<1=14.

Q71. Answer: B — 1

Topic: String Length + Modulo
ALGORITHM has 9 characters; 9%4=1.

Q72. Answer: B — S

Topic: String Reverse Access
Last index is 6; s[6] = S.

Q73. Answer: B — 2

Topic: String Character Count
PLACEMENT contains E twice.

Q74. Answer: B — 4

Topic: String Length + Right Shift
INTERVIEW has 9 characters; 9>>1=4.

Q75. Answer: A — N

Topic: String Conditional
Length is 7, so the first branch displays s[0]=N.

Q76. Answer: B — 70

Topic: Function Arithmetic
a becomes 10; b becomes 7; return 10×7=70.

Q77. Answer: C — 22

Topic: Function with Loop
1+(1+2+3+4+5+6)=22.

Q78. Answer: B — 45

Topic: Function with Shift
10<<2=40; 10>>1=5; total=45.

Q79. Answer: A — 18

Topic: Function with XOR
12 XOR 5 = 9; 9×2=18.

Q80. Answer: C — 18

Topic: Function Condition
14 is not divisible by 4, so 14+4=18.

Q81. Answer: D — 17

Topic: Function Digit Logic
3+8+6=17.

Q82. Answer: B — 18

Topic: Function Nested
g(4)=8 and g(5)=10; total=18.

Q83. Answer: A — 2

Topic: Function Modulo
49+3=52; 52%10=2.

Q84. Answer: C — 48

Topic: Function Loop Product
Multiply even values 2×4×6=48.

Q85. Answer: C — 15

Topic: Function Boolean
9 is odd and 6 is even, so return 9+6=15.

Q86. Answer: D — 14

Topic: Function Array Formula
5+3=8; 8 XOR 6 = 14.

Q87. Answer: C — 16

Topic: Function Recursive-style
f(7)=7+5+3+1=16.

Q88. Answer: C — 20

Topic: Function Shifted Return
13<<1=26; 13>>1=6; difference=20.

Q89. Answer: B — 8

Topic: Function with Multiple Updates
a=8; b=11; a=-3; return -3+11=8.

Q90. Answer: C — 30

Topic: Function with Loop and Shift
2×(1+2+3+4+5)=30.

Q91. Answer: C — 11

Topic: Mixed Array + Function
a[2]=16; 16>>1=8; 8+3=11.

Q92. Answer: A — 0

Topic: Mixed Loop + XOR
0^1=1, ^2=3, ^3=0, ^4=4, ^5=1.

Q93. Answer: B — 10

Topic: Mixed Loop + AND
x=15&13=13; 13&12=12; 12&11=8.

Q94. Answer: C — 41

Topic: Mixed Condition + Shift
19 is odd; 19<<1=38; +3=41.

Q95. Answer: B — 5

Topic: Mixed Nested Loop + Modulo
Pairs with sum divisible by 3: 5 pairs. Correct answer B.

Q96. Answer: C — 13

Topic: Mixed Array + Right Shift
41>>2=10; 18%5=3; total=13.

Q97. Answer: C — 31

Topic: Mixed Function + Logical
12>10 and 7<10, so (12<<1)+7 = 24+7=31.

Q98. Answer: C — 15

Topic: Mixed Loop + Array Update
Running cumulative values become [1,3,6,10,15].

Q99. Answer: B — 7

Topic: Mixed Function + Modulo + Shift
26%5=1; 1<<2=4; 26>>3=3; total=7.

Q100. Answer: C — 18

Topic: Mixed Nested Conditions
9>6 and 9–6=3, so the nested condition is true; 9+6+3=18.